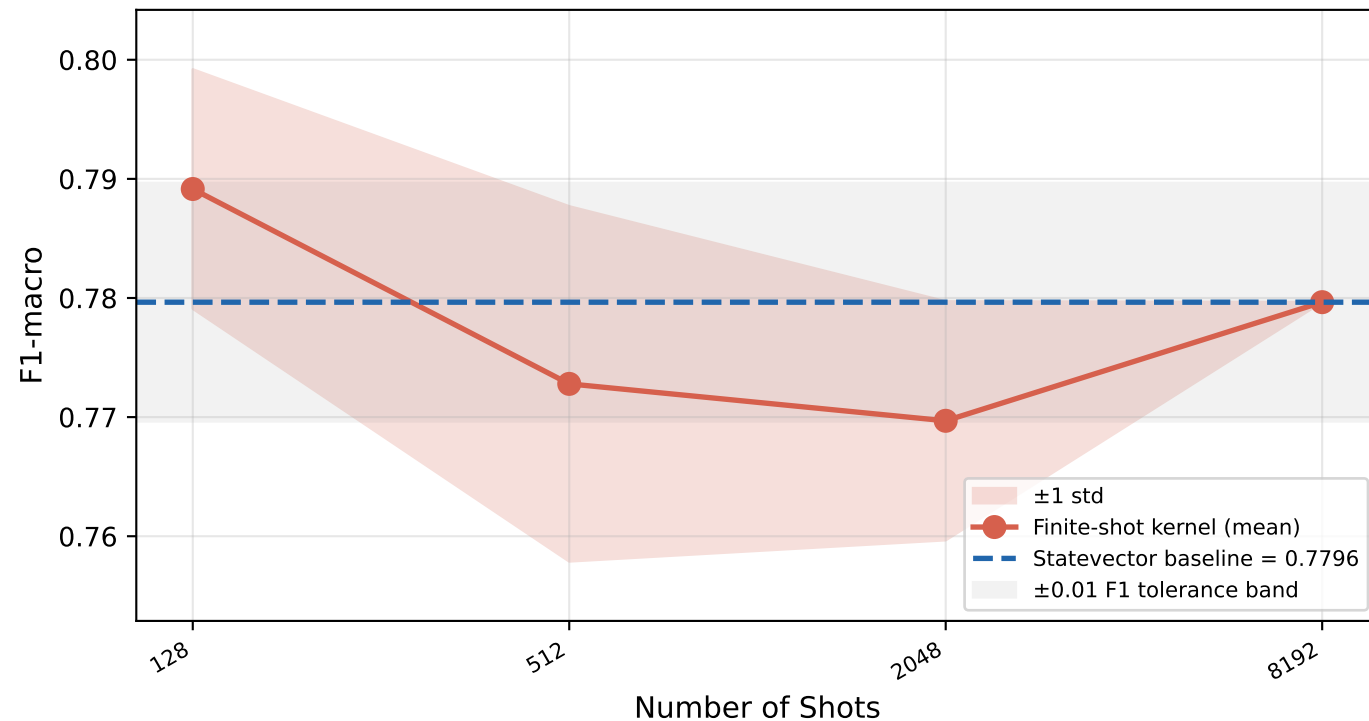


C2.5 — Finite-Shot Convergence Toward Statevector Kernel

(a) F1-macro convergence to statevector baseline



(b) Kernel convergence summary across shots

Shots	$\Delta F1$	Fro. Sim	Train MAE	Test MAE
128	-0.0095	0.9970	0.01518	0.01534
512	+0.0069	0.9991	0.00820	0.00820
2048	+0.0100	0.9996	0.00597	0.00587
8192	-0.0000	1.0000	0.00188	0.00185

Finite-Shot Convergence Summary

- SV baseline F1 = 0.7796
- Largest F1 drop = 0.0100 (shots=2048)
- Smallest $|\Delta F1|$ = 0.0000 (shots=8192)

• Train MAE 128→8192: 0.01518→0.00188

• Test MAE 128→8192: 0.01534→0.00185

Higher shots reduce kernel sampling error, while 8192 shots gives the closest match to the statevector baseline.